

Module 4

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Module 4

Dynamic Allocation

In Module 4, we will learn more about constructors, especially an important constructor known as the copy constructor, which is invoked automatically by the compiler whenever it needs to copy a class object. We also learn about destructors, which are functions that are automatically called when an object is about to go out of existence. Destructors give us a place to clean up after we're done with an object. We learn how to dynamically allocate and deallocate memory, and then learn how constructors, destructors, and memory allocation are intertwined. In this module we also see the overloading of the subscript operator [], which provides tremendous power. Finally, we put it all together in a large example, [DynamicIntArray](#).

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Learning Objectives

- Operators [new](#) and [delete](#)
- Destructors
- The copy constructor
- Overloading the assignment operator
- Overloading the subscript operator

Lectures

Presentation

- [4.1: Comments on Reading Assignments for Module 3](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-4-dot-1>)
- [4.2: Dynamically Allocating Memory: new and delete](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-4-dot-2>)
- [4.3: Destructors](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-4-dot-3>)
- [4.4: Initializations, Assignments, Copy Constructors and =](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-4-dot-4>)
- [4.5: DynamicIntArray Example](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-4-dot-5>)
- [4.6: delete vs. delete \[\]](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-4-dot-6>)

Reading

The reading for this Module is C review. Depending on your C background, you may be able to move through this material quickly.

Required Readings

- C++ *Primer Plus*, Chapter 12

Recommended Readings

- *Thinking in C++ Volume One*, pp. 459–496

Discussion

[Module 4 Discussion](#)

(https://onlinelearning.berkeley.edu/courses/2291660/discussion_topics/14351914)

Assignment

[Module 4: Short Answers 4](#)

(<https://onlinelearning.berkeley.edu/courses/2291660/assignments/33623150>)

[Module 4: Programming Assignment 4](#)

(<https://onlinelearning.berkeley.edu/courses/2291660/assignments/33623148>)

Source Code

In each module of this course, you will be required to submit a programming assignment. For each module, I have prepared sample source code for your reference. Please take a few minutes to download and install all of these programming examples. The files have been compressed into one zip. You will need a zip utility to decompress the file.

 [sources.zip](#)

[https://onlinelearning.berkeley.edu/courses/2291660/files/313890817/download?
wrap=1\)](https://onlinelearning.berkeley.edu/courses/2291660/files/313890817/download?wrap=1)